



Association of triglyceride glucose index with stroke: from two large cohort studies and Mendelian randomization analysis

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Introduction: The triglyceride glucose index (TyG) is associated with cardiovascular diseases; however, its association with stroke remains unclear. This study aimed to elucidate this relationship by examining two extensive cohort studies using two-sample Mendelian randomization (MR).

Methods: Using data from the 1999–2018 National Health and Nutrition Examination Survey (NHANES) and the Medical Information Mart for Intensive Care (MIMIC)-IV, the correlation between TyG (continuous and quartile) and stroke was examined using multivariate Cox regression models and sensitivity analyses. Two-sample MR was employed to establish causality between TyG and stroke using the inverse variance weighting method. Genome-wide association study catalog queries were performed for single nucleotide polymorphism-mapped genes, and the STRING platform used to assess protein interactions. Functional annotation and enrichment analyses were also conducted.

Results: From the NHANES and MIMIC-IV cohorts, we included 740 and 589 participants with stroke, respectively. After adjusting for covariates, TyG was linearly associated with the risk of stroke death (NHANES: hazard ratio [HR] 0.64, 95% CI: 0.41–0.99, $P=0.047$; Q3 vs. Q1, HR 0.62, 95% CI: 0.40–0.96, $P=0.033$; MIMIC-IV: HR 0.46, 95% CI: 0.27–0.80, $P=0.006$; Q3 vs. Q1, HR 0.32, 95% CI: 0.12–0.86; Q4 vs. Q1, HR 0.30, 95% CI: 0.10–0.89, $P=0.030$, P for trend = 0.017). Two-sample MR analysis showed genetic prediction supported a causal association between a higher TyG and a reduced risk of stroke (odds ratio 0.711, 95% CI: 0.641–0.788, $P=7.64e^{-11}$).

Conclusions: TyG was causally associated with a reduced risk of stroke. TyG is a critical factor for stroke risk management.

Keywords: Mendelian randomization, MIMIC-IV, NHANES, stroke, triglyceride-glucose index

Introduction

Stroke is a globally prevalent acute neurological disorder, posing a grave threat to life and leading to irreversible neurological impairment^[1,2]. Globally, more than 13.7 million people experience a stroke annually, accounting for more than half of all fatalities^[3]. Given the substantial impact of stroke on the life and health of a significant proportion of the population, there is an urgent need to establish timely and effective preventive interventions.

The triglyceride-glucose (TyG) index serves as a reliable biomarker of insulin resistance, reflecting insulin sensitivity. The TyG index is a practical tool for utilizing fasting blood glucose and routine biochemical test-derived glucose levels, thereby bypassing the conventional high insulin-normoglycemic clamp and homeostatic model assessment for insulin resistance (HOMA-IR) tests^[4]. Its simplicity, cost-effectiveness, and stability confer distinct advantages^[5]. However, limited research exists on the association between TyG and stroke risk and further investigations are warranted to establish a causal relationship.

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Sponsorships or competing interests that may be relevant to content are disclosed at the end of this article.

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International Journal of Surgery (2024) 110:5409–5416

Received 21 February 2024; Accepted 30 May 2024

Supplemental Digital Content is available for this article. Direct URL citations are provided in the HTML and PDF versions of this article on the journal's website, www.ijv.com/international-journal-of-surgery.

Published online 19 June 2024

<http://dx.doi.org/10.1097/JS9.0000000000001795>

Mendelian randomization (MR) analysis explores statistical methods of causality using genetic variation [single nucleotide polymorphisms (SNPs)] as an instrumental variable to identify causal associations by taking advantage of the fact that genetic variation is randomly assigned and is not subject to confounding factors^[6]. Thus, it appears feasible to use MR to assess the causal association between TyG and stroke.

This study investigated the association between TyG and stroke in two large cohort. Our findings validated through meticulous adjustments for variables and sensitivity analyses, confirm the stability of the observed association. Additionally, employing a two-sample MR enhances the causal understanding of this relationship. In summary, our study highlights the resilience of both cohorts and genetic factors, shedding light on the possible correlation between TyG and stroke.

Methods

Study design overview

This study comprised two main phases. In the initial phase, we conducted a comprehensive analysis of the correlation between the TyG index and stroke, accounting for various potential confounding factors. This analysis utilized data from both the National Health and Nutrition Examination Survey (NHANES) and Medical Information Mart for Intensive Care (MIMIC)-IV. Our findings were robustly validated through sensitivity analyses. Detailed data for extracting the two queues are shown in the flowcharts (Fig. 1).

In the subsequent phase, we extracted summary statistics from a genome-wide association study (GWAS) of TyG and stroke for two-sample MR analyses. Additionally, we performed molecular function and pathway analyses using genes identified through SNPs.

Two extensive observational cohort studies

NHANES Database Cohort All data for this study were sourced from the NHANES website (<https://wwwn.cdc.gov/nchs/nhanes/Default.aspx>), a comprehensive health screening and nutritional status survey encompassing adults and children in the United States. The dataset spans demographic, dietary, screening, laboratory, and questionnaire sections, covering 1999–2018. All study protocols were approved by the Ethics Review Board of the National Center for Health Statistics, and written informed consent was obtained from all participants before data collection commenced.

MIMIC-IV Database one of the authors (J.Y.A.) underwent formal training (record ID: 58,572,169). This study received ethical exemptions from the Massachusetts Institute of Technology and the Beth Israel Deaconess Medical Center, and no additional ethical approval was deemed necessary.

Data collection and definitions

NHANES Database Cohort We conducted a comprehensive analysis using the NHANES database, Covariates included age, sex, race, BMI, LDL, HDL, and total cholesterol (TC), systolic blood pressure (SBP), diastolic blood pressure (DBP), education level, poverty income ratio (PIR), smoking status, drinking status, physical activity, and diabetes. BMI categories followed WHO guidelines: normal or underweight ($<24.9 \text{ kg/m}^2$), overweight

HIGHLIGHTS

- In observational studies, the triglyceride glucose index has been associated with a reduced risk of death from stroke.
- The triglyceride glucose index was linearly associated with the risk of stroke death.
- Mendelian randomization analyses support a causal association between triglyceride glucose index and risk of stroke.
- Functional analyses also support that the stroke mechanism is mediated by lipid metabolism.

($25.0\text{--}29.9 \text{ kg/m}^2$), and obese ($\geq 30.0 \text{ kg/m}^2$). Education level was classified as high school degree or less, high school graduation, and college or higher. PIR was grouped into ≤ 1.3 , $1.31\text{--}3.50$, and > 3.50 . Smoking status was categorized as never, former, and current. Drinking status was recorded as a yes or no. Physical activity was assessed using the Global Physical Activity Questionnaire, distinguishing between inactive [< 600 metabolic equivalent (MET) minutes/week] and active (≥ 600 MET minutes/week). Diabetes was diagnosed based on Centers

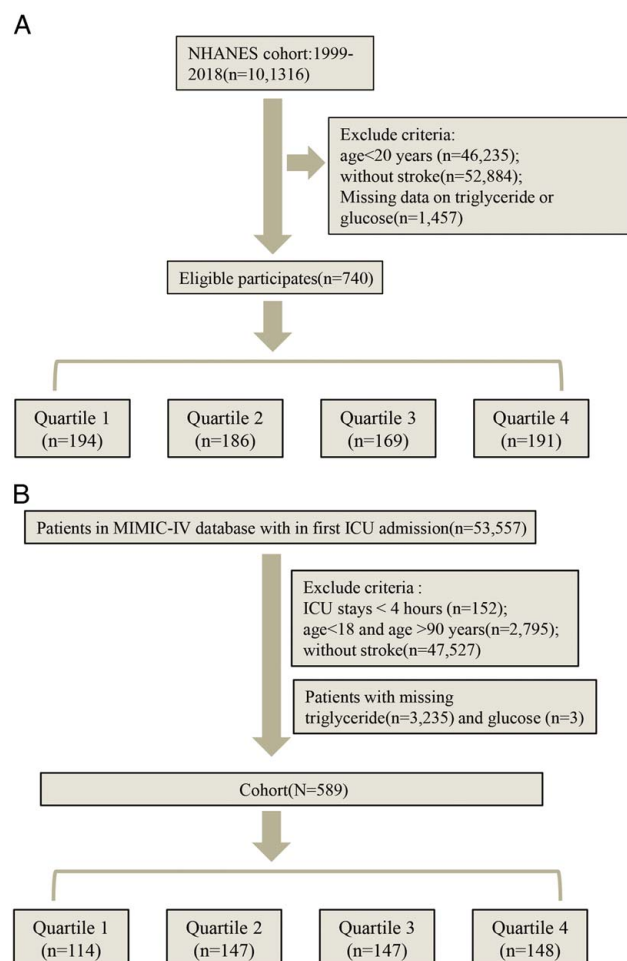


Figure 1. Flowchart of this study. A, NHANES cohort (1999–2018); B, MIMIC-IV cohort.

for Disease Control and Prevention (CDC) criteria (yes/no).

MIMIC Database Cohort As previously noted^[7], MIMIC-IV data were acquired with proper authorization, the demographic characteristics included age, sex, race, and weight. Laboratory test results included cholesterol level, white blood cell (WBC) count, red blood cell (RBC) count, SBP, and DBP. Documented comorbidities included congestive heart failure, peripheral vascular disease, hypertension, paralysis, vascular disease, and diabetes. Disease scores such as the Simplified Acute Physiology Score (SAPS) II, Acute Physiology Score (APS) III, Logistic Organ Dysfunction System (LODS), Glasgow Coma Scale (GCS), and Sequential Organ Failure Assessment (SOFA) were also considered. The initial recorded values were determined when the variables were documented more than once in the previous 24 h. The follow-up period commenced on admission and concluded at the occurrence of a specific endpoint of interest, providing crucial insights into patient progression.

The NHANES (1999–2018) and MIMIC-IV cohorts were systematically analyzed (Fig. 1). The missing values are detailed in Supplementary Table S1–Table S2 (Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>). Variables with more than 20% missing values were omitted, and multiple imputation techniques were applied to enhance the reliability of the results for variables with less than 20% missing values.

Triglyceride glucose (TyG) index definition

The TyG index is a key indicator calculated using the following formula: $\text{Ln} [\text{triglycerides (mg/dl)} \times \text{glucose (mg/dl)}^2]$.

Outcomes and follow-up

In the NHANES cohort (1999–2018), data on all-cause mortality were linked to the death linkage files until 31 December 2019. The follow-up time was computed from the examination date to either the date of death or the conclusion of the follow-up period (31 December 2019).

For the MIMIC-IV cohort, the follow-up period commenced more than 4 h after ICU admission and concluded with the occurrence of the specified outcome.

Statistical analysis

In the NHANES cohort (1999–2018), we used sample weights, pseudo-primary sampling units (PSUs) (sdmvpstu), and pseudostata (sdmvstrata) to accommodate a stratified multistage design across various sampling cycles. Following the NHANES guidelines, the sample weights were computed as 2/5WTMEC4YR for the years 1999–2002 and 1/5WTMEC4YR for the years 2003–2018 in subsequent analyses^[8,9].

Survey weighting was employed for the NHANES cohort (1999–2018) processing but not for the MIMIC-IV cohort.

TyG was computed as both a continuous and a categorical variable (quartiles: Q1–Q4) in both cohorts to evaluate its association with stroke. Mann–Whitney *U* or Kruskal–Wallis tests were used to assess the normal distribution. Continuous variables are presented as mean [standard error (SE)] or median [interquartile range (IQR)], and categorical variables as percentages (%). Analysis of variance (ANOVA) and χ^2 tests were used for group comparisons.

The Cox proportional risk model was used to determine the hazard ratios (HR) and 95% CI between TyG (continuous and

quartiles) and stroke-related all-cause mortality. Multivariate regression models adhering to the STROBE guidelines^[10] were applied to the NHANES cohort (Model 1: no adjustment; Model 2: adjusted for age, sex, race, and weight; Model 3: adjusted for age, sex, race, BMI, PIR, education, smoking, alcohol use, physical activity, diabetes, TC, LDL, HDL, SBP, and DBP). For the MIMIC-IV cohort, several models were included (Model 1: no adjustment; Model 2: adjusted for age, sex, race, and weight; Model 3: adjusted for age, sex, race, weight, cholesterol, WBC count, RBC count, SBP, DBP, congestive heart failure, peripheral vascular disease, hypertension, paralysis, vascular disease, diabetes, SAPS II, APS III, LODS, GCS, and SOFA).

Restricted cubic spline (RCS) analysis was used to explore the nonlinear relationship between TyG and all-cause stroke mortality.

The subgroup analyses focused on specific populations. In the NHANES cohort (1999–2018), attention was given to age (< 65 and ≥ 65 years), sex (female and male), race (non-Hispanic White and others), BMI (< 30.0 kg/m² and ≥ 30.0 kg/m²), smoking (yes/no), and diabetes (yes/no). In the MIMIC-IV cohort, analyses considered age (< 65 and ≥ 65 years), sex (male/female), race (Black, White, and other), congestive heart failure (yes/no), peripheral vascular disease (yes/no), hypertension, paralysis (yes/no), vascular disease (yes/no), diabetes (yes/no), and GCS^[3–15]. Likelihood ratio tests were used to assess stroke interactions with the stratification variables.

Sensitivity analysis

Sensitivity analysis was performed to enhance the stability of the results. In the NHANES cohort (1999–2018), individuals who experienced a stroke within 2 years were initially excluded to mitigate the risk of reverse causation. Given the potential impact of antihyperglycemic, antihyperlipidemic, and antihypertensive drugs on stroke risk, participants using these medications were excluded from the analysis. A similar approach was applied in the MIMIC-IV cohort, excluding patients using antihyperglycemic agents and antihyperlipidemic drugs, to bolster the robustness of the results.

Two-sample MR analysis

MR relied on three core assumptions to evaluate the causal association between exposure and outcomes: 1) SNPs chosen as instrumental variables exhibited strong associations with TyG (exposure), 2) genetic variants demonstrated no associations with other confounding factors, and 3) the impact of a genetic variant on stroke (outcome) was solely attributed to TyG.

Our study leveraged data from the Finnish cohort (finngen_R9_C_STROKE) within the FinnGen study^[11], a large-scale genomics initiative analyzing over 500 000 Finnish Biobank samples, to correlate genetic variations with health data and elucidate disease mechanisms and susceptibility. We utilized a summary-level GWAS dataset for stroke in the European population tested in 2023 ($n=311,635$; cases $n=39\ 818$; controls $n=271\ 817$). The validation cohort comprised publicly available data from the MEGASTROKE consortium encompassing 446 696 individuals of European ancestry (406 111 noncases and 40 585 stroke cases), including those with ischemic stroke, intracerebral hemorrhage, and strokes of unknown or undetermined types ($n=67\ 162$).

SNPs associated with the TyG index were selected from a previous GWAS ($P < 5 \times 10^{-8}$). This GWAS involved 273 368 participants aged 40–69 years without diabetes or lipid metabolism disorders^[12]. SNPs were excluded based on linkage disequilibrium ($R^2 < 0.01$, $kb = 10$ Mb), particularly those associated with triglycerides, glucose, and nonlipid/nonglycemic factors (including SBP, DBP, and BMI), to address potential horizontal pleiotropy ($P < 5 \times 10^{-8}$). Ultimately, 192 SNPs were selected as instrumental variables for the TyG index (Supplementary Table S3, Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>).

Instrumental variables selection and functional analysis

We meticulously filtered out nonpresent SNPs in the outcome GWAS through a rigorous series of steps and harmonized the exposure and outcome data. The identified SNPs were mapped to their corresponding genes using the GWAS Catalog (<https://www.ebi.ac.uk/gwas/>). To unravel the biological functions and pathway mechanisms associated with these SNPs, we conducted analyses using gene ontology and the Kyoto Encyclopedia of Genes and Genomes (KEGG). Interactions between different SNPs were explored using STRING (<https://string-db.org/>).

Univariate two-sample MR analysis

To assess the causal association between TyG and stroke, we leveraged GWAS data. The outcome estimate for the primary MR analysis was determined using inverse variance-weighting (IVW)^[13]. Complementary analyses included the weighted median^[14], MR-Egger^[15], and weighted mode. The MR-Egger intercept was employed for multivariate validity assessment, and Cochran's Q test was used to evaluate the heterogeneity among genetic variants.

Multivariate mendelian randomization (MVMR) analysis

We employed the IVW method as the primary analysis, adjusting for confounders such as alcohol consumption^[16], diabetes^[11], BMI^[17], and Apolipoprotein B (ApoB)^[18]. Additionally, we conducted co-adjustments for these confounders in sensitivity analyses. The MR analyses were performed using the TwoSampleMR, MR-PRESSO, and MVMR software packages in R (v.4.2.3; R Basis for Statistical Computing, Vienna, Austria). All P -values were two-sided, and statistical significance was defined as $P < 0.05$.

Data and resource availability

The MIMIC-IV cohort of this study is available from the Massachusetts Institute of Technology (MIT) and the Beth Israel Deaconess Medical Center (BIDMC), and the data are available to the authors upon reasonable request, with permission from MIT and BIDMC. The National Center for Health Statistics and Ethics Review Board approved the NHANES protocol, and all participants provided written informed consent.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Result

Baseline characteristics

In the NHANES cohort (1999–2018), the median age was 67.00 (54.00–77.00) years, with male participants ($n = 355$). In the MIMIC-IV cohort, the median age was 69.03 (58.25–78.70) years, with 52.12% male participants. The detailed baseline characteristics are presented in (Supplementary Table S4, Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>). The association between the TyG index and stroke was explored using TyG quartiles. In the NHANES cohort (1999–2018), a non-Hispanic White race, BMI ≥ 30 kg/m², and diabetes showed higher TyG indexes, with significant differences in LDL, HDL, and TC among TyG quartiles. In the MIMIC cohort (Supplementary Table S5, Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>), significant differences were observed in the TyG quartiles for congestive heart failure, diabetes mellitus, weight, APS III, LODS, SOFA, glucose, cholesterol, and WBC count.

Association between TyG index and stroke: results from two cohorts

The TyG index was evaluated as a continuous and categorical variable to assess its association with the risk of stroke-related mortality in the NHANES and MIMIC-IV cohorts. In the NHANES cohort (1999–2018), after comprehensive adjustment for potential confounders, a linear association was observed between the TyG index and stroke-related mortality (P for non-linearity = 0.164) (Figure S1). Notably, there was a 36% reduction in the risk of all-cause mortality for each unit increase in TyG level as a continuous variable. A 38% reduction in risk was observed for each unit increase in TyG level in the Q3 group. Trend analysis indicated a statistically significant association between increasing TyG levels and reduced all-cause mortality from stroke ($P = 0.015$) (Table 1). Sensitivity analysis, excluding stroke cases with follow-up within less than 2 years (Supplementary Table S6, Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>) and recent use of antihyperglycemic (Supplementary Table S7, Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>), antihyperlipidemic (Supplementary Table S8, Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>), and antihypertensive medications (Supplementary Table S9, Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>), consistently supported our findings.

In the MIMIC-IV cohort, the results mirrored those of the NHANES cohort (1999–2018), demonstrating a linear correlation (P for nonlinearity = 0.173) (Figure S1). Following adjustment for potential confounders, the TyG index as a continuous variable exhibited a negative association with the risk of all-cause mortality from stroke (HR 0.46, 95% CI: 0.27–0.80, $P = 0.006$). Categorically, it showed a negative association with mortality risk in the Q3 and Q4 groups (Q3, HR 0.32, 95% CI: 0.12–0.86, $P = 0.023$; Q4, HR 0.30, 95% CI: 0.10–0.89, $P = 0.030$). Consistent with the NHANES cohort (1999–2018), the risk of all-cause mortality from stroke increased progressively with increasing TyG index (P for trend = 0.017) (Table 1). Sensitivity analyses further supported our conclusions (Supplementary Table S10–S11, Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>).

Table 1
Association between TyG and stroke from the NHANES Cohort (1999–2018) and MIMIC-IV cohort.

NHANES Cohort ^a	TyG (continuous per 1 unit)	TyG quartile				P for trend
		Q1	Q2	Q3	Q4	
Model 1 ^b						
HR (95% CI)	0.99 (0.80–1.22)	Ref	0.95 (0.58–1.56)	0.97 (0.64–1.47)	1.00 (0.65–1.53)	
P	0.928		0.85	0.884	0.999	0.997
Model 2 ^c						
HR (95% CI)	0.99 (0.79–1.25)	Ref	1.19 (0.79–1.85)	0.82 (0.56–1.20)	1.05 (0.70–1.57)	
P	0.957		0.452	0.3	0.821	0.237
Model 3 ^d						
HR (95% CI)	0.64 (0.41–0.99)	Ref	1.08 (0.67–1.74)	0.62 (0.40–0.96)	0.70 (0.38–1.32)	
P	0.047		0.758	0.033	0.273	0.015
MIMIC-IV Cohort ^e						
Model 1 ^f						
HR (95% CI)	0.76 (0.50–1.15)	Ref	0.56 (0.23–1.37)	0.44 (0.18–1.08)	0.64 (0.29–1.42)	
P	0.197		0.204	0.074	0.276	0.291
Model 2 ^g						
HR (95% CI)	0.73 (0.47–1.13)	Ref	0.58 (0.23–1.47)	0.43 (0.17–1.07)	0.64 (0.28–1.46)	
P	0.16		0.254	0.071	0.285	0.247
Model 3 ^h						
HR (95% CI)	0.46 (0.27–0.80)	Ref	0.50 (0.18–1.36)	0.32 (0.12–0.86)	0.30 (0.10–0.89)	
P	0.006		0.17	0.023	0.03	0.017

^aNHANES Cohort:
^bModel 1: adjusted for none.
^cModel 2: adjusted for age, sex, race, and BMI.
^dModel 3: adjusted for age, sex, race, BMI, PIR, education, smoking, alcohol use, physical activity, diabetes, TC, LDL, HDL, SBP, and DBP.
^eMIMIC-IV Cohort:
^fModel 1: adjusted for none.
^gModel 2: adjusted for age, sex, race, and weight.
^hModel 3: adjusted for age, sex, race, weight, cholesterol, WBC count, RBC count, SBP, DBP, congestive heart failure, peripheral vascular disease, hypertension, paralysis, vascular disease, diabetes, SAPS II, APS III, LODS, GCS, and SOFA.
APS III, Acute Physiology Score III; DBP, diastolic blood pressure; GCS, Glasgow Coma Scale; HR, hazard ratio; HDL, high-density lipoprotein; HR, hazard ratio; LDL, low-density lipoprotein; LODS, Logistic Organ Dysfunction System; MIMIC, Medical Information Mart for Intensive Care; NHANES, National Health and Nutrition Examination Survey; PIR, poverty income ratio; SBP, systolic blood pressure; SAPS II, Simplified Acute Physiology Score II; SBP, systolic blood pressure; SOFA, Sequential Organ Failure Assessment; TC, total cholesterol; TyG, triglyceride glucose index; WBC, white blood cell; RBC, red blood cell.

Subgroup analysis

Subgroup analyses were conducted for both the stroke cohorts. In the NHANES cohort (1999–2018) (Fig. 2A), non-Hispanic White participants exhibited a higher stroke risk compared to other racial groups (HR 0.29, 95% CI: 0.14–0.61, $P=0.001$). Participants with a BMI ≥ 30.0 kg/m² (HR 0.27, 95% CI: 0.10–0.72, $P=0.009$) and those with diabetes (HR 0.37, 95% CI: 0.17–0.80, $P=0.011$) displayed a negative association between TyG index and all-cause mortality from stroke. A similar trend was observed in the MIMIC cohort (Fig. 2B), where the TyG index in participants aged ≥ 65 years (HR 0.49, 95% CI: 0.26–0.94, $P=0.031$, P for interaction = 0.026) and those with paraplegia (HR 0.26, 95% CI: 0.07–0.99, $P=0.048$) was negatively linked to the risk of all-cause mortality from stroke. This association persisted in both cohorts.

Two-sample Mendelian analysis of TyG and stroke, with validation

To further validate our observational findings across the two cohorts, we conducted a one-way two-sample MR to assess the causal relationship between TyG index and stroke risk. The results revealed a significant association, indicating that an elevated TyG index was associated with an increased risk of stroke (OR 0.711, 95% CI: 0.641–0.788, $P=7.64 \times 10^{-11}$) (Fig. 3).

Sensitivity analyses demonstrated no horizontal pleiotropy (intercept = -0.003 , $P=0.064$) or heterogeneity (Cochran's $Q=102.425$, $P=0.931$) for the selected SNPs (Supplementary Table S12, Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>). Additional complementary MR methods, including MR-Egger ($P=0.011$), weighted median ($P=0.005$), and weighted mode ($P=0.019$) consistently supported our primary findings. To further strengthen our conclusions, we validated our results using the MEGASTROKE Consortium's stroke GWAS pooled data, which confirmed a similar phenomenon (OR 0.783, 95% CI: 0.698–0.878, $P=2.95 \times 10^{-05}$) (Supplementary Table S13, Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>). Moreover, we mapped common SNPs ($n=113$) from both cohorts to protein-coding genes (Supplementary Figure S2A–S2C, Supplemental Digital Content 1, <http://links.lww.com/JS9/C792>). Subsequent investigations using the STRING protein interaction database as well as gene ontology and KEGG analyses revealed significant enrichment of these genes in biological functions such as lipid homeostasis, triglyceride homeostasis, and regulatory pathways of cholesterol metabolism. To address potential confounding factors, we performed multivariate two-sample MR analyses after adjusting for stroke risk factors (alcohol consumption, diabetes, BMI, and ApoB)

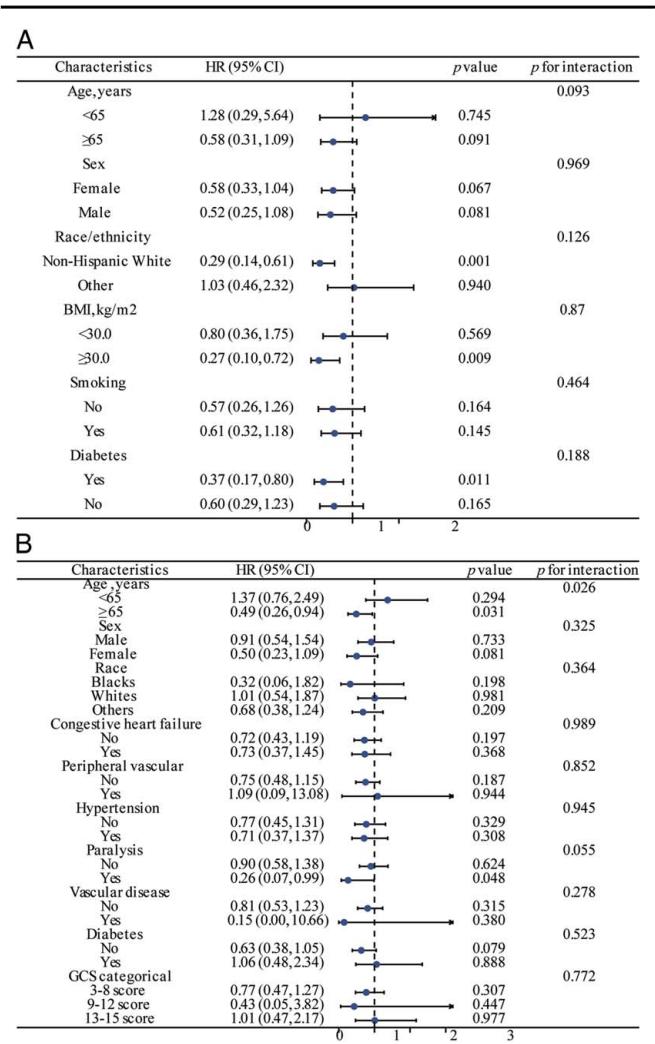


Figure 2. Subgroup analysis of TyG index and stroke. A, NHANES cohort (1999–2018). B, MIMIC- IV cohort. HR, hazard ratio; GCS, Glasgow Coma Scale.

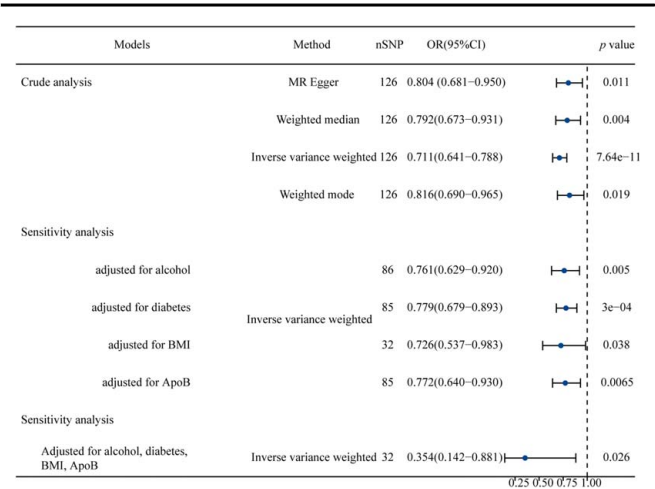


Figure 3. Mendelian randomization to determine the causal association between TyG index and stroke. OR, odd ratio; SNP, single nucleotide polymorphisms.

(Fig. 3). The results remained stable in both single-adjusted and fully adjusted risk factor models.

Discussion

A large number of previous studies have confirmed a strong association between the TyG index and diseases such as diabetes, atherosclerotic cardiovascular disease, hypertension, and heart failure^[19–21]. In NHANES cohort (1999–2018), the risk of death from stroke decreased by 36% for each unit increase in the TyG index, and in the Q3 group, the risk of death from stroke decreased by 38%. In the MIMIC-IV cohort, the risk of death from stroke was reduced by 54% for each unit increase in the TyG index, and the risk of death was reduced by 68% in the Q3 group and 70% in the Q4 group.

- The possible reasons for this are as follows:
- (1) High or low levels of triglycerides and glucose can be detrimental to health, and both can lead to worsening of disease. Low levels of glucose increase adrenaline levels, which can further lead to vasoconstriction and platelet buildup, promoting cardiovascular and cerebrovascular events^[22]. Low triglyceride levels are associated with recurrent ischemia and higher mortality in patients with acute coronary syndrome^[23]. A U-shaped association of the TyG index with mortality in CVD patients with diabetes mellitus or prediabetes mellitus (all-cause mortality risk HR 0.47; cardiovascular mortality risk HR 0.25)^[24].
 - (2) We observed that the majority of participants included in the NHANES and MIMIC cohorts were typically overweight, from BMI and weight data. In the study by Hou *et al.*^[25], mortality was significantly lower in overweight/obese stroke patients than in stroke patients with normal/low BMI. This may be due to the obesity-stroke paradox. Adipose tissue secretes soluble tumor necrosis factor (TNF)-α receptors and, therefore, neutralizes the biological effects of TNF-α. Some studies have explained that obese patients may receive early preventive measures from clinicians and earlier treatment with antithrombotics, antihypertensives, and statins^[26–28]. However, in our study, we performed sensitivity analyses (including baseline use of hypoglycemic, lipid-lowering, and antihypertensive medications), and our results did not support this explanation. Another argument that seems more reasonable is that the obesity paradox occurs more frequently in patients with insulin resistance; most patients with insulin resistance have obesity and are overweight, and the TyG index is more responsive to a disturbed insulin resistance state^[29]. In patients with acute ischemic stroke undergoing intravenous thrombolysis, a low percentage of visceral abdominal fat is associated with good outcomes^[30]. This finding suggests that obesity has unexpected protective effects.
- MR, a technique for exploring causal associations which assumes that the allele of interest is randomly and uniformly distributed in the population of interest, is similar to a randomized controlled experiment and can effectively overcome the limitations of observational studies^[31]. In our two-sample MR analysis, a causal association between the TyG index and reduced risk of stroke, which is consistent with our observational study and was validated using other GWAS data. This finding significantly strengthened the authenticity and validity of the results.

To the best of our knowledge, this is also the first study to elaborate on the relationship between the TyG index and stroke risk.

Strengths and limitations

The strengths of our study are as follows: (1) detailed and comprehensive analysis of two independent cohorts strengthened our findings through a series of sensitivity analyses; and (2) we further addressed the limitations of traditional observational studies through a MR approach and employed multivariate two-sample MR to consolidate our conclusions, with multiple methods to test for pleiotropy and heterogeneity.

In addition, there were limitations to this study. First, as an observational cohort study, we only had BMI data from the NHANES cohort and weight data from the MIMIC cohort (height data were missing in large numbers), and the NHANES data were obtained based on questionnaire self-reporting from the population, which is not ideal for diagnostic accuracy. Second, the obesity paradox is a concerning problem, and visceral obesity, other factors, and selection bias in the population may affect our conclusions. Third, the MR analysis was confined to a European population; generalizations of other populations deserve our consideration, and the robustness of the study is still relatively limited.

Conclusions

Our findings suggest that there is an association between the TyG index and stroke risk reduction, and that our findings have implications for stroke risk control not only in clinical treatment but also in prevention. We look forward to large-scale, multi-ethnic prospective studies addressing this issue in the future.

Ethical approval

All studies had been approved by a relevant ethical review board and participants had given informed consent. Ethical approval was not required because of the public characteristics of the data of GWAS.

Consent

Informed consent was not required for this study

Source of funding

This study was supported by grants from the National Natural Science Foundation of China (No. 82260378), High-end Talent Program for Science, Technology and Innovation (No. G3423), and Natural Science Foundation of Jiangxi Provincial Science and Technology Department (No. 20232ACB206019).

Author contribution

All authors contributed significantly to, and are in agreement with, the content of the manuscript. Z.Y.: conceptualization; J.Y.A. and S.J.: data curation and resources; C.P. and C.J.H.: formal analysis and software; Z.Y.: funding acquisition; Z.Y.Y., L.J.W., and C.J.H.: investigation; C.J.H. and C.S.Q.: methodology; Z.Y.: supervision; J.Y.A. and S.J.: validation; Z.Y.Y. and

L.J.W.: visualization; J.Y.A. and S.J.: writing – original draft; Z.Y.: project administration and writing – review and editing and ZY is the guarantor of this work and, as such, had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Conflicts of interest disclosure

This study did not involve any research conducted on human or animal participants. The authors declare no conflict of interest.

Research registration unique identifying number (UIN)

1. Name of the registry: not applicable.
2. Unique identifying number or registration ID: not applicable.
3. Hyperlink to your specific registration (must be publicly accessible and will be checked): not applicable.

Guarantor

Zhang Yan.

Data availability statement

The datasets that were obtained in this study can be made available by the corresponding author upon reasonable request.

Provenance and peer review

Not applicable.

Acknowledgement

Genetic association estimates were obtained from a genome-wide association meta-analysis of the MRC-IEU study and the FinnGen consortium. The authors thank all the investigators for sharing these data. We also thank “easyMR” for some code support.

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